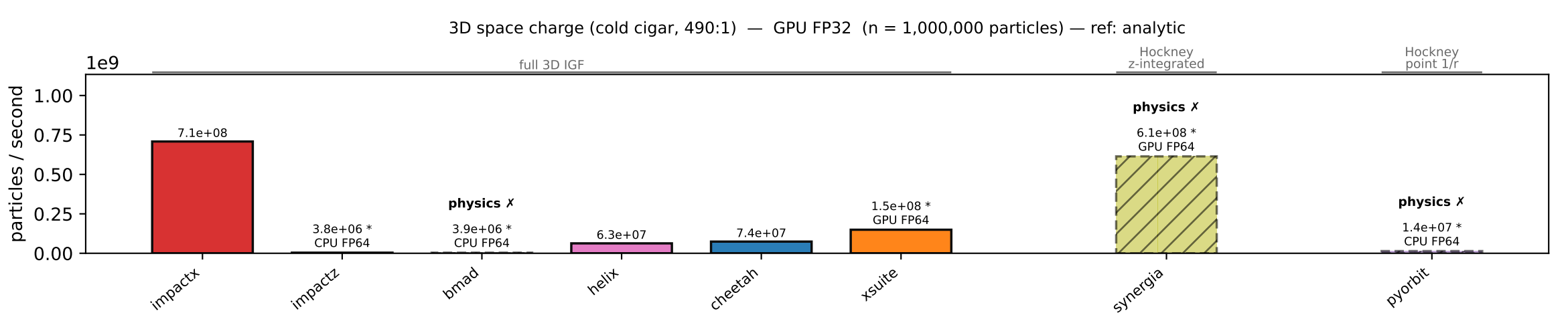

'physics ✗' = converged but out of tolerance

** next-best shown: impactz, bmad, pyorbit = CPU FP64 (no GPU); xsuite, synergia = GPU FP64 (no FP32)*

kernel over a mesh cell: IGF = integrated in 3D; z-integrated = along z only, point-sampled in x,y; point 1/r = sampled at cell centres

versions: impactx 26.08 · impactz V2.7.7 · bmad 20260904-1 · helix v1.10.1 · cheetah 0.8.2 · xsuite 0.53.4 · synergia v2024.04-209-g795b99f39 · pyorbit 96c8def